Example 5.9 Find the value of 'a' such that $PQ = QR$ where P, Q, and R are the points whose coordinates are (6, –1), (1, 3) and (a, 8) respectively.

Solution

Given P (6, –1), Q (1, 3) and R (a, 8)

$$PQ = \sqrt{(1-6)^2+(3+1)^2} = \sqrt{(-5)^2+(4)^2} = \sqrt{41}$$

$$QR = \sqrt{(a-1)^2+(8-3)^2} = \sqrt{(a-1)^2+(5)^2}$$

Given $PQ = QR$

Therefore $\sqrt{41} = \sqrt{(a-1)^2+(5)^2}$

$41 = (a-1)^2 + 25$ [Squaring both sides]

$(a-1)^2 + 25 = 41$

$(a-1)^2 = 41 - 25$

$(a-1)^2 = 16$

$(a-1) = \pm 4$ [taking square root on both sides]

$a = 1 \pm 4$

$a = 1 + 4$ or $a = 1 - 4$

$a = 5$ or $a = -3$

Example 5.10 Let $A(2, 2)$, $B(8, -4)$ be two given points in a plane. If a point P lies on the X- axis (in positive side), and divides AB in the ratio 1: 2, then find the coordinates of P.

Solution

Given points are $A(2, 2)$ and $B(8, -4)$ and let $P = (x, 0)$ [P lies on x axis]

By the distance formula

$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$AP = \sqrt{(x-2)^2+(0-2)^2} = \sqrt{x^2-4x+4+4} = \sqrt{x^2-4x+8}$$

$$BP = \sqrt{(x-8)^2+(0+4)^2} = \sqrt{x^2-16x+64+16} = \sqrt{x^2-16x+80}$$

Given $AP : PB = 1 : 2$

i.e. $\frac{AP}{BP} = \frac{1}{2}$ $(\because BP = PB)$

$2AP = BP$

squaring on both sides,

$4AP^2 = BP^2$

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